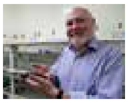

**Biologists 'transfer' a memory**
Biologists transfered a memory between two marine snails, creating an artificial memory, by injecting RNA. **https://dgit.in/Jun18-B30-1**

**Croco-dolphin?**
A 180-million year old fossil could shed light on how ancient crocodiles evolved into dolphin-like animals. **https://dgit.in/Jun18-B30-2**

more thrust than the Saturn V. However, the Saturn could still put in more tons into orbit. The SLS series of vehicles will also be used for the early manned missions to Mars. The SLS has a number of planned upgrades. The launch capacity is expected to go up from 95,000 kg to 130,000 kg to low Earth orbit (LEO) over the course of these upgrades. The SLS is being developed as a partnership between NASA, Orbital ATK and Boeing. Over at Europe, the European Space Agency is in the process of developing the Ariane 6, which can place up to 20,000 kg in LEO. The rocket will reduce the costs of launches as compared to the Ariane 5, and allow more launches. The first flight is expected to take place around 2020. Mitsubishi is developing the H3 rocket for JAXA, which will be a medium lift launch vehicle. The debut flight is expected to take place in 2020. China has a prolific fleet of Long March launch vehicles, and to reach its space ambitions, it plans to introduce the Long March 8 to its lineup. The rocket is tweaked to launch satellites in the sun synchronous orbit. The Long March 9, will have the capability of injecting 1,40,000 kgs into LEO, and the first flight is expected to take place in 2028. Jeff Bezos' Blue Origin is also expected to settle into a regular launch schedule, with the New Glenn being able to place 45,000 kg into LEO. The debut New Glenn launch is expected to take place in 2020. ISRO plans to introduce a 15,000 kg to LEO capacity rocket, known as the Unified Launch Vehicle. The low capacity is actually by design, a way to cut down costs, and to cater to a new market of CubeSat launches. This smaller launch vehicle will be capable of regularly launching microsatellites, nanosatellites and picosatellites. To increase its launch capacity, ISRO also plans to open up a third launchpad at its facility in Sriharikota. The largest capacity of all, would of course be that of the Big Falcon Rocket (BFR) by SpaceX. The debut launch is expected to take place in 2022. The BFR will be capable of deploying 150,000 kg into LEO in the reusable configuration, and a staggering 250,000 kg in the expendable configuration. In the expendable configuration, the lower stages are not reused after the launch.

## THE MOON

In the latter half of this year, ISRO has scheduled its second mission to the moon. There are three components to the mission, an orbiter, a soft lander and a rover. The purpose of the mission is to conduct a mineralogical study of the lunar surface. Towards the end of 2018, CNSA, the Chinese space agency has also planned a mission to the moon. The Chang'e 4 mission will also consist of an orbiter, a lander and a rover. If successful, the mission will be the first to land on the far side of the Moon. A year later, China will attempt a Lunar sample return mission, the Chang'e 5. Around the same time, JAXA has also planned the Selene-2 follow up mission, which will also consist of an orbiter, a lander and a rover. In 2021, JAXA plans the Smart Lander for Investigating Moon (SLIM) mission, an autonomous spacecraft that will decide where it wants to land, instead of a convenient landing spot. The mission hopes to land in a precise location, with 200 times more resolution than the Apollo missions. SpaceX has agreed to take two mysterious tourists on an orbit around the Moon, and the launch was originally scheduled for 2018. However, Elon Musk has indicated that the it will be the Big Falcon that will carry the Dragon with the first Lunar tourists inside. Whoever commissioned the journey would have to wait till at least 2022. Virgin Galactic, or Blue Origin may start flying tourists to space before that time though. Boeing and Lockheed Martin can expected to enthusiastically participate in the space tourist race as well. China and Russia have some tentative plans for manned missions to the moon, but it is unclear if these will be realised within the next decade.

The space agencies of a number of countries are collaborating on a long term project that will pave the way for future manned and unmanned missions spanning the solar system. CSA, JAXA, ESA, Roscosmos and NASA are all partners in the Lunar Orbital Platform-Gateway, also known as the Deep Space Gateway. This space station will orbit the Moon, in the space between the Earth and the Moon. It will be used as a staging area for further missions to the Moon and Mars. The experiments, technology demonstrations and testing needed for the equipment used in robotic and manned missions, will be tested in the deep space environment of this space station.

A Big Falcon Rocket deploying its Payload